

The Long-term Effects of Bank Bailouts on Corporate Financing Policies

Nobuyuki Kanazawa*

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Abstract

This paper examines the long-term effect of bank bailouts on borrower firms' financing policies using the Japanese banking crisis and the subsequent bank bailouts in the 1990s. I study this effect by exploiting the variation in firms' reliance on the bailed-out banks a year before the bailouts. The results show that firms that had a relationship with the bailed-out banks experienced a persistent increase in the long-term debt to asset ratio and a persistent reduction in the cash to asset ratio. These findings suggest a lasting impact of the bank bailout on borrower firms' financing policies. Additionally, I find that borrower firms are more likely to sustain their banking relationship with the same main banks from the pre-bailout periods. The finding is consistent with the existing literature, which emphasizes the role of a close relationship with banks in alleviating the financial constraints of firms.

Keywords: Capital injection; Banking crisis; Corporate borrowing; Cash holdings; Bank relationships.

JEL Classification: E44; G21; G31; G32.

*Lecturer, Faculty of Economics, Soka University, 1-236, Tangi-machi, Hachioji-shi, Tokyo, 192-8577, Japan. Email: nkanazawa@soka.ac.jp.

1 Introduction

The immediate impacts and aftermath of past financial crises have highlighted the importance of the health of financial institutions for the overall economy, including sectors that are far removed from banking. When balance sheets of banks worsens, borrower firms face credit supply contraction and reduce employment and investment ([Chodorow-Reich \(2014\)](#), [Almeida et al. \(2011\)](#), and [Bentolila et al. \(2018\)](#)).¹ To limit the impact of such balance sheet deterioration of banks, governments in the wake of financial crises often recapitalize troubled banks, despite the deep public unpopularity of such recapitalization. Recent empirical literature on the effects of bank bailouts shows that sufficient capital injections into troubled banks are indeed effective at stimulating the banks to increase credit supply, which then helps increase the borrower firms' investment ([Giannetti and Simonov \(2013\)](#) and [Kasahara et al. \(2019\)](#)).² While previous work has highlighted the short-term benefit of bank bailouts on borrower firms' corporate policies, few studies have examined its long-term effects.

This paper estimates the long-term impacts of bank bailouts on borrower firms' financing policies. Specifically, I exploit the Japanese banking crisis and subsequent bank bailouts in the 1990s to examine the long-term impact of the bailouts on borrower firms' cash holding behaviors and financial debt positions. Soon after the banking crisis of 1997, the Japanese government conducted consecutive bank recapitalizations on March 1998 (1.8 trillion yen) and March 1999 (7.5 trillion yen). I exploit the variation in the strength of the pre-existing financial relationships of the firms with the bailed-out banks to understand the long-term impact of the bank bailouts on borrower firms.

This paper uses the “exposure to injection” measure compiled by [Giannetti and Simonov \(2013\)](#) as an indicator for the strength of firms' financial ties with bailed-out banks. This measure calculates the proportion of loans that each firm received from any of the bailed-out banks in the year preceding the recapitalizations. Using this exposure measure as the treatment indicator, I use the two-way fixed effects difference-in-differences estimator to estimate the long-term effects of bank bailouts on the borrower firms' financing policies. The estimated results reveal two patterns. First, the borrower firms' long-term debt as a share of the total assets increased significantly after the bank recapitalization and remained persistently

¹Other studies that show real effects of credit supply contraction include [Alfaro et al. \(2021\)](#), [Cingano et al. \(2016\)](#), [Peek and Rosengren \(1997\)](#), [Kashyap and Stein \(2000\)](#), [Amiti and Weinstein \(2018\)](#), and [Acharya et al. \(2018\)](#).

²In addition, capital injections are found to be effective in the growth of bank-dependent firms amid the crises ([Laeven and Valencia \(2013\)](#)).

high even after two decades. Second, the borrower firms' cash to asset ratio declined immediately after the recapitalizations and remained persistently low. These findings suggest a permanent shift in the financing policies of the borrower firms due to the bank recapitalizations.³ The baseline results are robust to different specifications, such as the inclusion of a variety of controls, use of the propensity score matching difference-in-difference estimator, and the removal of zombie firms from the sample.

Finally, I discuss an additional finding that helps understand a potential mechanism of the persistent shift in the borrower firms' financing policies. I find that the borrower firms are more likely to sustain banking relationships with the same main banks from the pre-recapitalization periods. This finding suggests that the firms reduced cash holding and increased bank loans partly because of their sustained (and thus closer) relationship with the banks, which is consistent with existing literature that discusses the banks' role in mitigating the financial constraint. [Hu et al. \(2016\)](#) reported that firms that have more intense relationships with banks have lower levels of cash. [Chauhan et al. \(2018\)](#) used Indian data to show that firms with bank-appointed directors have lower cash holdings.⁴

This paper contributes to the literature on the effects of bank bailouts. [Giannetti and Simonov \(2013\)](#) examined the Japanese bank bailouts in 1998, 1999, and 2002 and showed that the first two recapitalizations were successful at stimulating bailed-out banks to extend loans to their borrower firms. Furthermore, [Giannetti and Simonov \(2013\)](#) show that having received a greater amount of loans, these borrower firms increased investment in the year following the recapitalizations. [Kasahara et al. \(2019\)](#) found a similarly positive effect of bank bailouts using the same Japanese bailout episodes.⁵ While the existing literature mostly focuses on the short-term effects, this paper examines the long-term effects of bank bailouts on the banks' client firms.

This paper also contributes to the literature on the determinants of corporate cash holding. According to [Keynes \(1936\)](#), there are two reasons for corporate cash holdings, which are the reduction of transaction cost and precautionary motive.⁶ In particular, the precautionary demand for cash theory states that firms accumulate cash to protect themselves against

³I could not detect any distinctive long-term effects for other key variables such as investment.

⁴However, their findings are in contrast to [Nakajima and Sasaki \(2016\)](#) and [Shikimi \(2019\)](#) who find that bank-dependent firms hold more cash.

⁵In addition, [Veronesi and Zingales \(2010\)](#) and [Bayazitova and Shivdasani \(2012\)](#) examine the impacts of the US government's equity infusion on the value of banks' financial claims.

⁶Other theories include the agency costs of free cash flow ([Jensen \(1986\)](#); [Ozkan and Ozkan \(2004\)](#)) and the corporate governance structure ([Dittmar et al. \(2003\)](#); [Harford et al. \(2008\)](#); [Pinkowitz et al. \(2006\)](#)).

adverse cash flow shocks that increase the external financing cost. Without sufficient cash in hand, firms that are hit by adverse cash flow shocks could be forced to forgo profitable investment opportunities. Empirical papers such as [Opler et al. \(1999\)](#), [Kahle and Stulz \(2013\)](#), and [Duchin \(2010\)](#) generally find the existence of this precautionary motive.

The finding in this paper is consistent with the precautionary motive for cash holding, especially in the context of financial constraints. Previous studies have documented that financially constrained firms hold higher levels of cash (e.g. [Almeida et al. \(2004\)](#), [Faulkender and Wang \(2006\)](#), [Acharya et al. \(2007\)](#), [Denis and Sibilkov \(2010\)](#), and [Harford et al. \(2014\)](#)). In particular, banks play an important role in alleviating the borrower firms' financial constraints. [Sasaki and Suzuki \(2019\)](#) find that worsening bank health causes firms to increase cash holding out of their cash flows. Similarly, this paper finds that bank bailouts induce borrower firms to maintain long-lasting relationships with their main banks. The sustained relationship with the banks would reduce informational asymmetry and help ease the external financing cost for borrower firms, which would enable a greater amount of bank borrowing.

The rest of the paper proceeds as follows: Section 2 discusses the background of the Japanese banking crisis of the 1990s and the following successive bank bailouts. Section 3 delineates the empirical approach employed in this paper. Section 4 explains the data used in the study. Section 5 summarizes the main results. Section 6 presents various robustness checks of the main results. Section 7 investigates the potential mechanism that drives the main results. Finally, Section 8 concludes this paper.

2 Background of the Japanese banking crisis and bailouts in the 1990s

This section briefly introduces the Japanese banking crisis and the following bank bailouts in the 1990s. After the stock market reached its highest level in late 1989 in Japan, the housing and real estate bubble burst, which gradually made the financial system unstable. In 1991, small financial institutions started to fail. In response, the deposit insurance system was activated for the first time in Japanese history. However, despite these efforts, small financial institutions continued to fail, and the non-performing loans in many banks continued to pile up.

From around 1995, concerns about the systemic implications of the failure of non-bank companies specializing in housing finance spread among the public. Eventually, in 1996,

seven housing finance companies went bankrupt, and public funds were injected to deal with some of these bankruptcies. Consequently, the government and the ruling party came under severe criticism over the use of public funds.

In November 1997, the largest bank in northern Japan (*Hokkaido Takushoku Bank*) and the fourth largest securities company in Japan (*Yamaichi Securities*) went bankrupt. Then, in the fall of 1998, two major banks, the Long-Term Credit Bank of Japan and the Nippon Credit Bank, also went bankrupt. In response to these events, the government decided to inject public capital into the major banks in rapid succession from 1998 to 1999.

In the first round of recapitalization in 1998, the government injected 1.8 trillion yen with most of the banks in the program receiving 100 billion yen, and some smaller banks receiving between 20 and 60 billion yen. On average, this amounted to 1.9 percent of banks' risk-weighted assets. Mirroring the harsh public criticism over the use of public funds to help financial institutions, the size of recapitalization was limited. In addition, the capital injection was designed to avoid revealing any signals about the financial institutions' health to the market. According to [Giannetti and Simonov \(2013\)](#), the size of injection to each bank was linked only to the size of the banks.

The second recapitalization was conducted in March 1999 to help banks that remained undercapitalized because of the small size of the first recapitalization. The size of injection in the second recapitalization was 7.5 trillion yen with each bank receiving different amounts between 200 and 1,000 billion yen. On average, each bank received about 5.1 percent of bank risk-weighted assets.

As [Kasahara et al. \(2019\)](#) noted, these injections were conducted by the government's purchase of preferred stock or subordinated debt and helped many banks to achieve the capital standard required under the 1988 Basel Accord. I mainly exploit the second recapitalization for the construction of the treatment variable because of its larger size.⁷ The analysis using the first recapitalization delivers similar results, as the indicators of firms'

⁷The sample in our study can be classified into four groups: 1) firms that are never affected, 2) firms that are only affected by the first round of recapitalization, 3) firms that are only affected by the second round of recapitalization, and 4) firms that are affected by both the first and second rounds of recapitalization. The baseline analysis uses the first and second groups (firms that are never affected and firms that are affected by the first round only) as the control group and the third and fourth groups (firms that are affected by the second round only and firms that are affected by both rounds) as the treatment group. In [Appendix 9.3.3](#), I check the robustness of the baseline results to a different definition of the control and treatment groups, where the control group consisting of only the firms that are never treated and the treatment group consisting of only the firms that are affected by both rounds of recapitalizations. The baseline results are robust to this alternative definition of control and treatment groups.

exposure to the first and second recapitalizations are highly correlated.

3 Empirical approach

This paper examines the long-term consequence of bank bailout on borrower firms' financing policies. To examine this, I employ the two-way fixed effects difference-in-differences estimator. The key variable is each firm's capital injection exposure denoted as Exposure Intervention_{*i*}, which is defined by [Giannetti and Simonov \(2013\)](#) as the proportion of loans that firm *i* received from any of the banks benefiting from the second recapitalization in the year preceding the recapitalization. Specifically, Exposure Intervention_{*i*} is calculated as follows:

$$\text{Exposure Intervention}_i = \sum_k \text{loans}_{i,k} / \text{total amount of loans}_i, \quad (1)$$

where loans_{*i,k*} denotes the amount of loans that firm *i* received from the bailed-out bank *k* in the year preceding the recapitalization, and total amount of loans_{*i*} is the total amount of the loan that firm *i* received from any bank in the year preceding the recapitalization. Using this variable, I define T_i as a treatment indicator that takes the value of 1 if Exposure Intervention_{*i*} > 0 and 0 otherwise.^{8,9} I estimate the model as follows:

$$Y_t = \sum_{t=1990}^{2016} \beta_t (\text{Year}_t \times T_i) + \psi X_{i,t} + \text{Firm}_i + \text{Industry}_h \times \text{Year}_t + u_{i,t}, \quad (2)$$

where Y_t is a dependent variable, and $X_{i,t}$ is the time-variant control variables. Because the recapitalizations mainly targeted larger banks, which tend to extend loans to large firms, one concern might be that the baseline results are simply capturing different progressions of financing policies of firms of different sizes. To address this concern, I included in $X_{i,t}$ the log of total assets and log of employment as control variables. In estimating equation 2, β_{1990} to β_{1996} are expected to be insignificant if the parallel trend assumption is satisfied. I set the year 1995 as the base year. The short and long-term effects of capital injection on the dependent variables are captured by β_{1997} to β_{2016} , as the first recapitalization was

⁸I prefer using the indicator variable rather than directly using Exposure Intervention_{*i*} in the regression because the estimated effects of the bank bailout exhibit some non-linear patterns as shown in [Appendix 9.1](#). For the long-term debt over asset ratio, for example, the effect of the bailout is the strongest among the bottom 25 percentile of firms (firms that have the weakest link to the bailed-out banks). Even when Exposure Intervention_{*i*} is directly used instead of T_i as a treatment variable, the results are qualitatively similar. However, the use of Exposure Intervention_{*i*} directly in the regression increases the standard errors.

⁹Approximately, 58% of the firms in the sample are *treated* under this specification.

implemented in March 1998, which falls in the 1997 fiscal year in Japan. $Firm_i$ represents the firm-level fixed effects. The model also includes the interaction term between industry fixed effects and year fixed effects to absorb any unobserved industry time-varying corporate characteristics. The standard errors are clustered at the firm level.

4 Data

The main data source is the Nikkei NEEDS FinancialQUEST. Using the NEEDS database, I collect accounting and loan information for all listed companies in Japan for the period 1990-2016. The main variables of our analysis are the long-term debt over total assets and cash over the total assets. Long-term financial debt is calculated by summing up the long-term debt and the current portion of the long-term debt. Cash is specified as the cash holding plus deposit. Other variables used in the analysis, such as total assets and the number of employees, are all retrieved from the Nikkei NEEDS data. Specifically, the NEEDS data contains the amount of loans and the banks' names for all listed companies in Japan.

I also use the dataset compiled by [Giannetti and Simonov \(2013\)](#) for measuring the strength of each firm's pre-existing relationship with the bailed-out banks from the pre-bailout period as defined in equation 1. After I construct the exposure to intervention measure using [Giannetti and Simonov \(2013\)](#), I restrict the sample to firms that existed during the bailout periods in the 1990s and were recorded in the [Giannetti and Simonov \(2013\)](#)'s dataset. Table 1 lists the main variables that I use in the analysis.

Table 1: Descriptive Statistics

	Observations	Mean	SD	Median	1st	99th
Long-term debt/TA	44,958	0.109	0.120	0.071	0	0.518
Cash/TA	44,958	0.133	0.094	0.113	0.011	0.458
log(total assets)	44,958	11.089	1.563	10.901	7.868	15.474
log(employees)	44,943	7.309	1.467	7.154	4.127	11.218
Interest Coverage ratio	44,284	0.679	6.016	0.061	0	10.342
Corporate tax/TA	44,958	0.021	0.019	0.016	0	0.084
Investment	44,625	0.081	1.172	0.023	-0.273	0.898

Notes. The table reports descriptive statistics for the main variables. Long-term debt/TA is the long-term financial debt of firm i in time t divided by the firm's total assets in year t . Cash/TA is the cash of firm i in year t divided by the firm's total assets in year t . Log(total assets) and log(employees) are the natural logarithm of total assets and the number of employees, respectively. I remove observations with negative asset values. Interest Rate Coverage is the interest rate coverage ratio of firm i in year t , defined as earnings before interest and taxes divided by interest expense of the firm in year t . Corporate tax/TA is the corporate tax payment by firm i in year t divided by the firm's total assets in year t . Investment is the growth rate of fixed assets of firm i between t and $t + 1$.

5 Results of the baseline model

To investigate the long-term effects of the bank bailout, I estimate equation 2 by first setting the long-term debt over total assets ratio as the dependent variable, and second, by setting cash over total assets ratio as the dependent variable. Figure 1 shows the results.

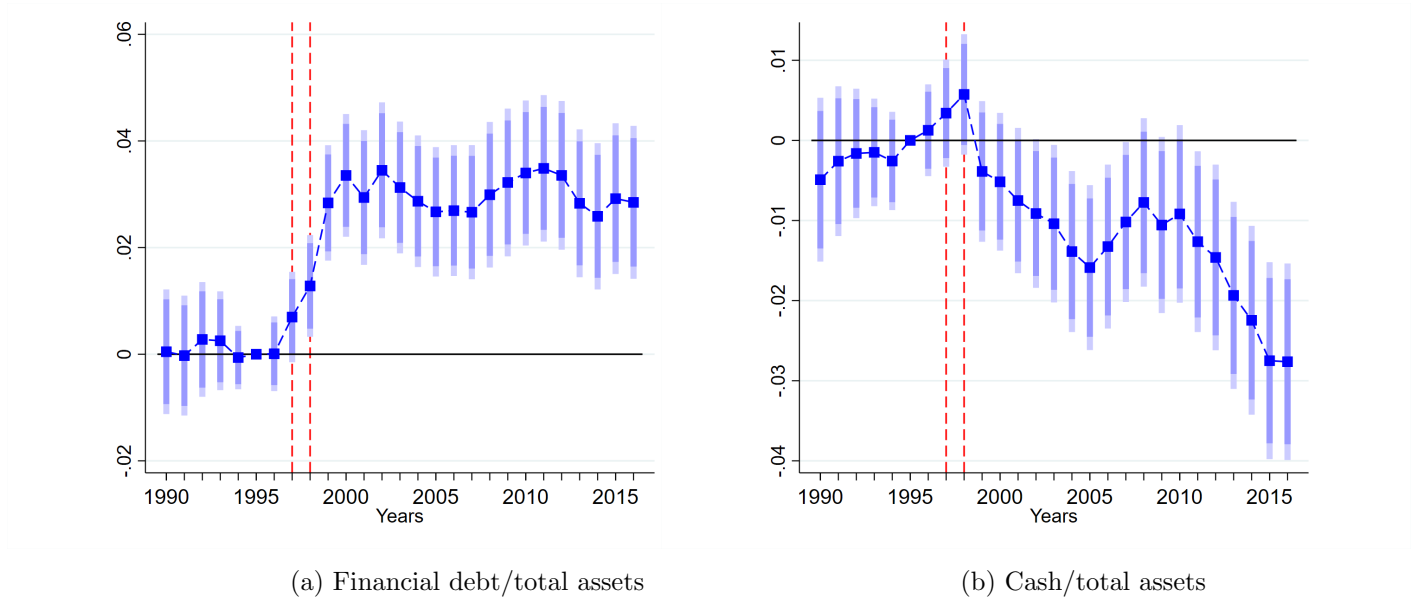
Panel (a) of Figure 1 shows $\hat{\beta}_t$'s for $t = 1990, \dots, 2016$ from equation 2 when the dependent variable is specified as long-term financial debt over total assets. The dark shaded area represents 90%, and the lightly shaded area represents 95% firm-cluster-robust confidence intervals, respectively. The dotted vertical lines indicate the first and second recapitalizations in 1997 and 1998.¹⁰ The long-term debt over asset ratio rises sharply during and immediately after the two recapitalizations. On average, the debt to asset ratio increases by 3 percentage points for the borrower firms due to the bank bailout. Furthermore, the level of debt over asset ratio remains persistently high years after the recapitalizations, indicating a permanent increase in the debt to asset ratio for these firms. The estimated increase in the debt to asset ratio after the recapitalizations are all statistically significant. Importantly, $\hat{\beta}_t$'s during the pre-recapitalizations period ($t = 1990, \dots, 1996$) are all statistically insignificant, suggesting the validity of the parallel trend assumption in this analysis.

Panel (b) of Figure 1 shows $\hat{\beta}_t$'s when the dependent variable is specified as the cash over total assets ratio. The cash to asset ratio immediately drops following the second recapitalization and keeps declining long after the recapitalization. The cash to asset ratio is about 2 percentage points lower for the borrower firms by 2013 than it is for the non-borrower firms. The estimated results are statistically significant in most years after the recapitalization. Again, the $\hat{\beta}_t$'s during the pre-recapitalization period are all statistically insignificant, suggesting the legitimacy of the key parallel trend assumption.

Together, Figure 1 suggests that the firms that had a pre-existing relationship with the bailed-out banks experienced a permanent shift in their financing policies. Put differently, the firms that suffered credit contraction because of their lack of pre-existing relationships with the bailed-out banks decreased their reliance on banks as a source of financing and increased the use of internal financing in the long-run.

¹⁰In addition to the second recapitalization, I show the timing of the first recapitalization in the Figures because many of the firms that were exposed to the second recapitalization were also exposed to the first. Therefore, the effect of the bailout could show up soon after the first recapitalization.

Figure 1: Baseline results



Notes. The figure shows the coefficients on the interaction term between the treatment indicator and the year fixed effect, $\hat{\beta}_t$ for $t = 1990 \dots 2016$, using equation 2, when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of employees). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

6 Robustness check

In this section, I check the robustness of the results of the baseline as follows. First, I control additional firm-specific time-varying variables. Second, I conduct a placebo test using the alternative treatment indicator. Third, I employ the propensity score matching difference-in-difference estimator. The baseline result is robust to all of these changes. In Appendix 9.3, I show the results of three additional robustness checks, which are the addition of firms' market valuation as a control¹¹, the removal of zombie firms from the sample, and the use of a different definition of control and treatment groups. The baseline result is robust to these different specifications as well.

6.1 Adding control variables

To check the robustness of the baseline results, I first add several additional control variables to the baseline model. Specifically, I add the following eight variables to $X_{i,t}$ in equation 2: interest coverage ratio, total liquid liability, firm age and age squared, corporate income tax payment over total assets, a lag of investment, and a lag of employee growth rate. Firms' age reflects the firms' growth opportunities and the uncertainty they face, which affects their external financing costs and need for cash holding. The interest coverage ratio, liquid liability, tax to asset ratio, investment, and employee growth rate are added to capture the firms' needs to accumulate cash, as these items often require payment in cash. Figure 2 shows the results.

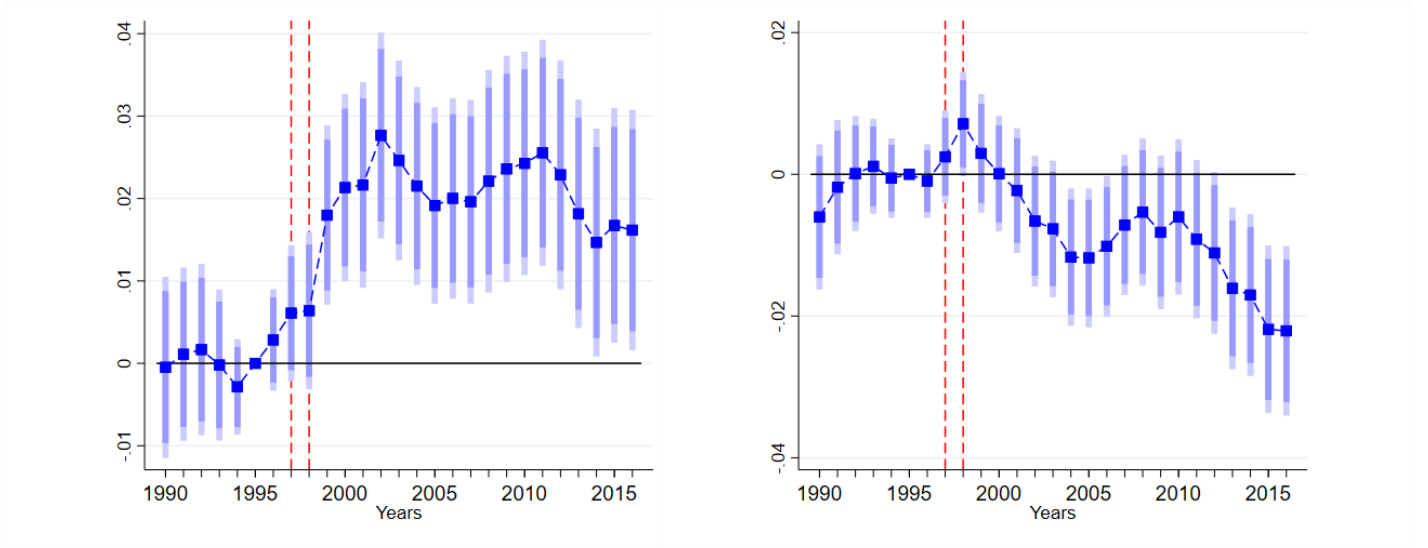
Panel (a) of Figure 2 shows the estimated effects of bailouts on the long-term debt to asset ratio. Although the magnitude of the increase in the debt to asset ratio slightly scales down compared to the baseline result, the pattern of the significant and persistent increase in the debt to asset ratio is still evident. Panel (b) shows the estimated effects on the cash to asset ratio. Again, the inclusion of the additional control variables does not make any meaningful difference to the estimated result. Overall, the baseline results are robust to the inclusion of the added control variables.

6.2 Placebo test

In this section, I conduct a placebo test by estimating equation 2 using a placebo treatment indicator. To construct the placebo treatment indicator, I exploit the third recapitalization

¹¹I relegated the results that control the market capitalization to the appendix because the data on market capitalization could not be obtained for as many as half the firms, in some years between 1990 and 1996 (pre-intervention periods).

Figure 2: Additional controls



(a) Financial debt/total assets

(b) Cash/total assets

Notes. The figure shows $\hat{\beta}_t$ for $t = 1990 \dots 2016$ using equation 2 when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets, log of the number of employees, interest coverage ratio, total liquid liability, firm age and age squared, corporate income tax over total assets, a lagged investment, and a lagged employee growth rate). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

episode in June 2003, when the Japanese government nationalized the Resona bank by injecting approximately 2 trillion yen through preferred and common shares. The nature of the third recapitalization is different from the previous two recapitalizations. The third recapitalization was triggered when the auditing corporation in charge of the Resona bank informed the bank in May 2003 that it would only allow deferred tax assets for three years instead of five years, which had been the guideline for auditing major banks at that time. This meant that the capital adequacy ratio of the Resona bank for the fiscal year ending March 2003 would fall below the domestic banks' standard of 4%. The news caught the public by surprise because the Resona bank was a newly established bank as a result of a merger of four banks in March the same year. I exploit this round of recapitalization for constructing a placebo treatment indicator as [Giannetti and Simonov \(2013\)](#) showed that, unlike the previous two recapitalizations, it did not lead the bailed-out bank to extend loans to borrower firms. Thus, we should not expect any long-term effects of this bailout on the financing policies of the borrower firms.

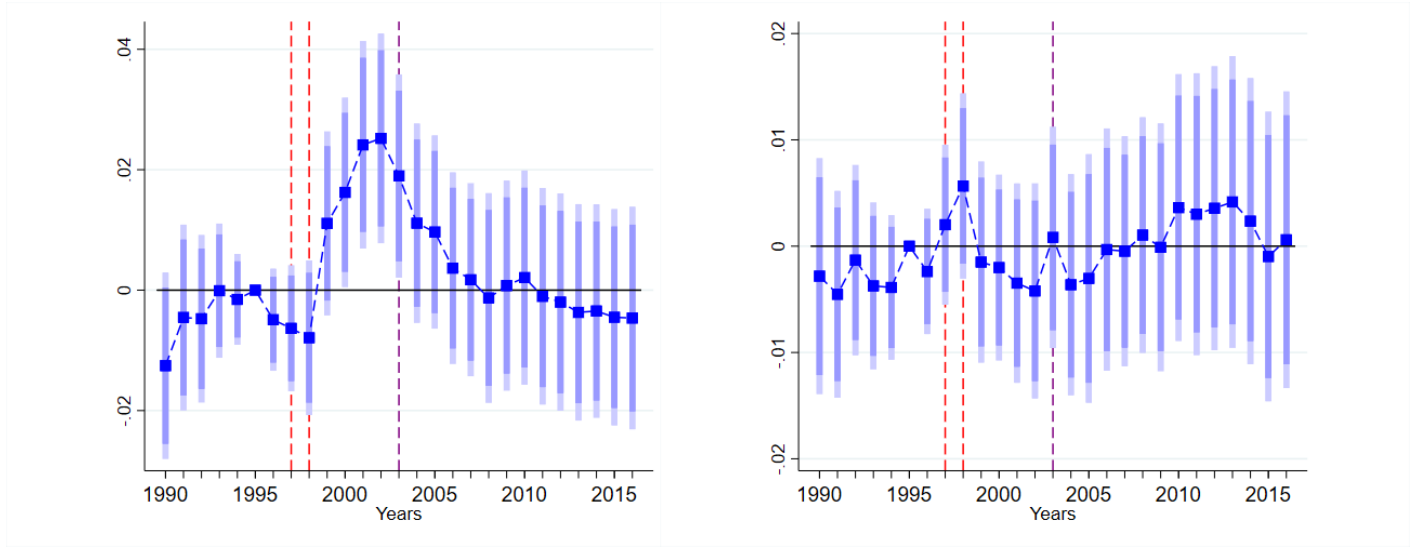
I define the placebo treatment indicator, T_i^{placebo} , as an indicator for the firm i that had borrowed from the Resona bank in the year preceding the 2002 recapitalization. Firms that meet this criterion constitute about 16% of the entire sample of firms. I estimate regression equation 2 using T_i^{placebo} as the treatment variable instead of T_i . Figure 3 shows the results.

Panel (a) of Figure 3 shows the estimated results of the placebo regression when the dependent variable is the long-term debt to asset ratio. The additional dotted vertical line indicates the timing of the third recapitalization in 2003. The debt to asset ratio climbs up even prior to the third recapitalization. This indicates that leading up to the formation of the Resona bank in 2003, the banks that were to be merged as the Resona bank tried to expand the loans, especially to small and medium enterprises, as documented in [Maeda \(2015\)](#).¹² As [Giannetti and Simonov \(2013\)](#) report, there is no discernible increase in debt to asset ratio for the borrower firms as a result of the 2003 recapitalization.

Panel (b) shows the results for the cash to asset ratio. The declining pattern of the cash to asset ratio that was evident in the baseline result is no longer observable in the placebo regression. Overall, the placebo regressions failed to replicate the baseline results.

¹²In Figure 7 in Appendix 9.2, I show that the share of outstanding loans from the four banks that eventually constitute the Resona holdings (*Yamato* bank, *Asahi* Bank, *Kinki-Osaka* Bank, and *Nara* Bank) compared to the overall outstanding loans from all domestic banks increases substantially before the merger.

Figure 3: Placebo test

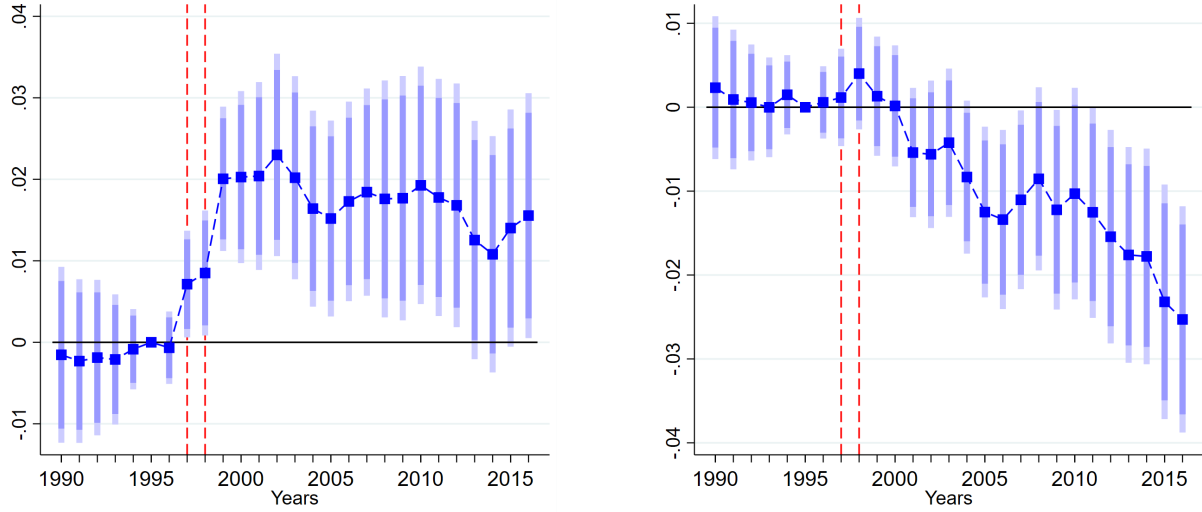


(a) Financial debt/total assets

(b) Cash/total assets

Notes. The figure shows $\hat{\beta}_t$ for $t = 1990 \dots 2016$ using equation 2 when the placebo treatment indicator is used. The dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of the number of employees). The first two recapitalizations occurred in 1997 and in 1998, followed by the third in 2003. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

Figure 4: Propensity Score Matching DiD estimator



(a) Financial debt/total assets

(b) Cash/total assets

Notes. The figure shows $\hat{\beta}_t$ for $t = 1990 \dots 2016$ using equation 2 weighted by the propensity scores when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and log of total assets, and log of the number of employees. The propensity score is calculated based on the outcome variables (debt to asset and cash to asset ratios), the log of total assets, and the log of the number of employees between 1990 and 1996. The bank bailouts occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

6.3 Propensity score matching difference-in-differences

As an additional robustness check, I use the propensity score matching difference-in-differences (PSM DiD) estimator. Combining the difference-in-differences model with propensity score matching enables further adjustment of the comparability between the treated (borrower firms) and untreated (non-borrower firms) groups based on the observable characteristics during the pre-intervention period. The propensity scores are calculated based on the debt to assets and the cash to asset ratios, log of total assets, and log of the number of employees between 1990 and 1996. The estimation is conducted using the kernel matching method. I then estimate the propensity score weighted version of equation 2 (Heckman et al. (1997)).

The results, presented in Figure 4, demonstrate significant effects of the bank bailouts consistent with the baseline results (though slightly smaller in size for the debt to asset ratio).¹³

¹³I also conducted the PSM-DiD using nearest neighbor matching. The results are insensitive to the different choices of matching algorithms.

7 Mechanism

In this section, I explore potential mechanisms for the long-term effects of bank bailouts on corporate financing policies. First, I examine if the probability of becoming unlisted from the Tokyo Stock Exchange differs across the treated and untreated groups. Second, I investigate if the bailout has affected the firms' probability of switching main banks.

7.1 Effects on the probability of exit

A potentially relevant mechanism that could explain the baseline results is the shift in the composition of firms in the treated and untreated groups. For example, because the untreated firms did not receive extra bank loans during the crisis, they may have exited from the stock exchange more frequently than the treated firms. In such a case, a greater number of unhealthy firms might have disappeared from the untreated group, which helps explain why the untreated firms that remain in the sample accumulate less debt and more cash.

To investigate the relevance of this composition effect, I calculate the predicted probabilities of becoming unlisted from the stock exchange for the treated and untreated firms. Specifically, I run the following regression:

$$I_{i,2017}^{\text{unlisted}} = \mathbf{f}(\alpha + \gamma T_i + \Phi W_{i,1997\dots 2004} + \epsilon_{i,2017}), \quad (3)$$

where $I_{i,2017}^{\text{unlisted}}$ is an indicator for firms that became unlisted by 2017. I include a number of control variables in $W_{i,1997\dots 2004}$ to examine the effect of the bailout on the eventual unlisting that is not caused by the initial difference in firms' characteristics across treated and untreated groups. $W_{i,1997\dots 2004}$ include industry fixed effects, current and two lags of zombie index, log of market capitalization, investment, interest coverage ratio, financial debt over total assets, and cash over total assets from 1997 to 2004. I estimate equation 3 using the linear probability model and the probit model. Table 2 shows the result. I find that the probability of becoming unlisted from the stock exchange by 2017 is slightly lower for the treated firms, but the effect is statistically indistinguishable from zero. The finding suggests that the main findings are unlikely to be caused by the mechanical shift in the compositions of firms due to un-listing from the stock market.

Table 2: Probability of Unlisted

	(1) Linear Prob. Model	(2) Probit Model
$d(\text{Prob. Unlist})/dT$	-0.033 (0.023)	-0.033 (0.022)
Number of obs.	1391	1362
R^2	0.130	0.055

Notes. The table reports the marginal effects of injection exposure on the probability of being unlisted in 2017 using the linear probability model in column (1) and the Probit model in column (2). Both models include industry fixed effects, current and two lags of zombie index, log of market capitalization, investment, interest coverage ratio, financial debt over total assets, and cash over total assets from 1997 to 2004 as control variables. Robust standard errors in parentheses. R^2 for Probit model is a pseudo- R^2 .

7.2 Effects on banking relationship

Another potential mechanism involves the firms' relationships with the main banks. Previous empirical papers have identified that financial constraint is an important determinant of firms' cash holding behavior¹⁴, and the relationship with the main banks plays a pivotal role in alleviating the financial constraint. Because the treated firms were given extra loans following the bank bailouts, these firms might be induced to maintain their pre-existing relationships with the same banks, which could help mitigate the informational asymmetry and reduce financial constraints. To investigate this mechanism, I estimate the conditional probabilities of sustaining the relationship with top 1 and top 3 banks from pre-bailout periods for each year following the recapitalization. Specifically, I estimate the following linear probability model:

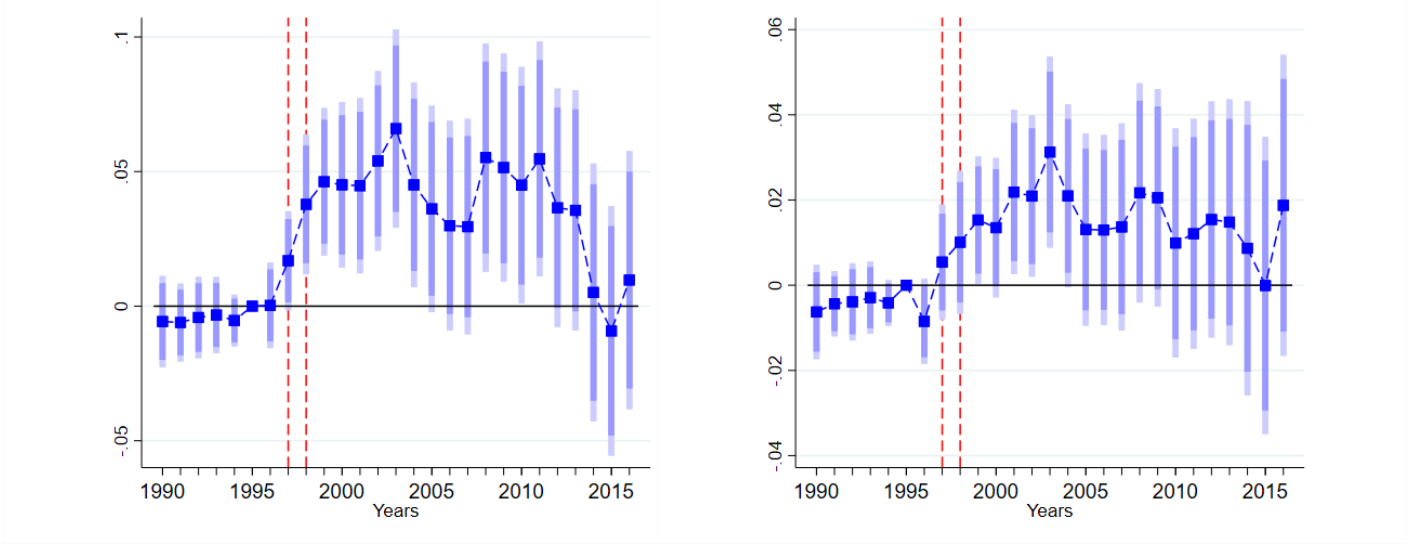
$$I_{i,t}^{\text{same bank}} = \sum_{t=1990}^{2016} \beta_t (\text{Year}_t \times T_i) + \psi X_{i,t} + \text{Firm}_i + \text{Industry}_h \times \text{Year}_t + u_{i,t}, \quad (4)$$

where $I_{i,t}^{\text{same bank}}$ is an indicator that takes the value of 1 if the firms i 's main bank (the bank from whom the firm received the largest amount of loans) in year t is the same as their main banks between 1990 and 1996. Figure 5 shows the results.

Panel (a) of Figure 5 shows the conditional probability of maintaining the same main

¹⁴For example, Almeida et al. (2004), Faulkender and Wang (2006), Acharya et al. (2007), Denis and Sibilkov (2010), and Harford et al. (2014).

Figure 5: Probability of sustaining banking relationship



(a) Top main bank

(b) Top three banks

Notes. The figure shows the estimated coefficients on the interaction term between the treatment indicator and the year fixed effects, $\hat{\beta}_t$ for $t = 1990 \dots 2016$, using equation 4 when the dependent variable is specified as the indicator for maintaining the same top 1 bank as the pre-bailout period in Panel (a) and the indicator for maintaining the same top 3 banks as the pre-bailout period in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of the number of employees). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

bank in year t as the pre-bailouts period. The firms that had a relationship with the bailed-out banks are approximately 5% more likely to keep using the same main bank as the pre-bailout period than the untreated firms. The effect is statistically significant and long-lived, returning to zero only after the year 2014. Panel (b) shows the conditional probability that the firms maintain the relationship with the top 3 banks from the pre-bailout period. Although the size of the effects weakens, the overall trend is similar, meaning that the treated firms are more likely to maintain the financial relationship with the same top 3 banks from the pre-bailout period. The result is consistent with the existing literature that emphasizes the role of banks in alleviating financial constraints for firms. The finding suggests that the bailout allowed banks to extend a greater amount of loans to borrower firms during the crisis, which had an additional effect of inducing the borrower firms to maintain the relationship with those banks for a longer period of time. Consequently, the borrower firms might have experienced a reduction in the external financing costs, which would prompt them to increase the debt and reduce cash holding.

8 Conclusion

This paper examined the long-run effects of bank bailouts on the financing policies of firms that had a pre-existing relationship with the bailed-out banks. The paper finds that firms that had a relationship with the bailed-out banks experienced a persistent increase in the long-term debt to asset ratio and a persistent reduction in the cash to asset ratio. These findings suggest a long-run shift in the firms' financing policies due to the bailouts. I also find that firms that had a relationship with the bailed-out banks had a higher probability of maintaining the relationship with the same main banks from the pre-bailout period. This finding is consistent with the existing literature, which emphasizes the role of a close relationship with banks in alleviating firms' financial constraints.

9 Appendix

9.1 Strength of effects

In this section, I estimate the potentially different effects of the bailout on corporate financing policies, depending on the strength of the firms' pre-existing connections with the bailed-out banks. To do this, I define T_i^q , which takes the value of 1 if the firm i belongs to the q -th quartile of the Exposure Intervention $_i^j > 0$ with $q = 1$ being the bottom 25 percentile (the weakest exposure to the bailed-out banks among firms that had pre-existing relationships with the bailed-out banks) and $q = 4$ denoting the top 25 percentile (the strongest exposure). I estimate the following model:

$$Y_t = \sum_{q=1}^4 \sum_{t=1990}^{2016} \beta_t^q (\text{Year}_t \times T_i^q) + \psi X_{i,t} + \text{Firm}_i + \text{Industry}_h \times \text{Year}_t + u_{i,t}, \quad (5)$$

If the effect of bailout increases according to the strength of the relationships with the bailed-out banks, we expect the absolute values of β_t^q to be greater as q increases. Figure 6 shows the results.

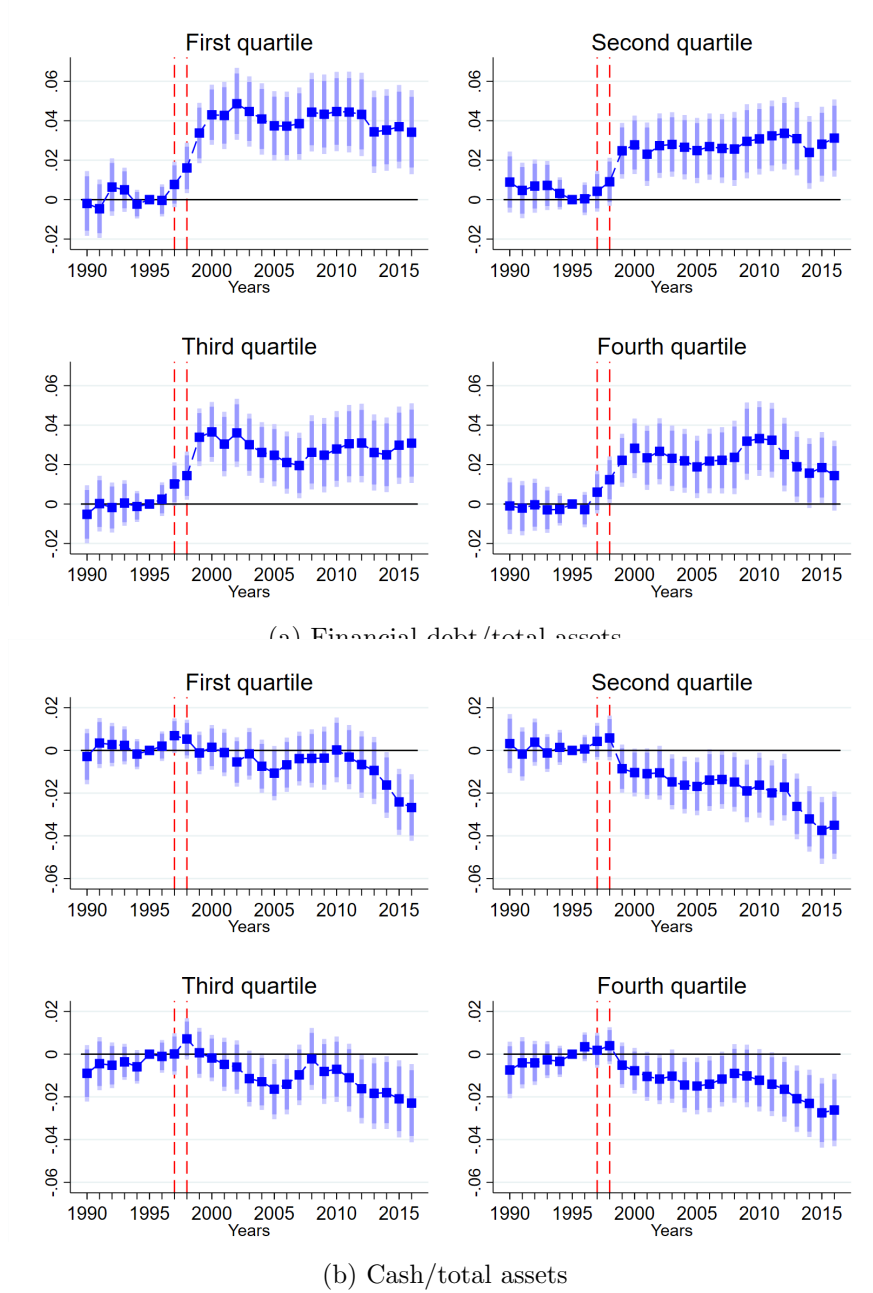
Panel (a) of Figure 6 shows the effects of the bailout on the long-term debt over asset ratio for different quartiles of firms according to the intensities of the firms' connections with the bailed-out banks. First, the effects of bailouts on the debt over asset ratio are visible across all quartiles. Second, the strength of the effect does not seem to correlate well with the intensities of the firms' connection to the bailed-out banks. The firms with the weakest exposure to the bailed-out banks ($q = 1$) experienced the strongest increase in the debt to asset ratio in the post-recapitalization period. The finding suggests that the strength of the bailout's effect on the debt over asset ratio does not linearly depend on the intensities of the firms' exposure to the bailout.

Panel (b) of Figure 6 shows the treatment effects on the cash over asset ratio for different quartiles of firms. Again, the effects of the bailout on the cash holding are significant across all groups of firms. The strength of the bailout's effect on cash holding appears to be weakly correlated with the intensities of the firms' connection with the bailed-out banks. The effects generally become stronger as the exposure to bailed-out banks increases.

9.2 Outstanding loans from Resona Holdings

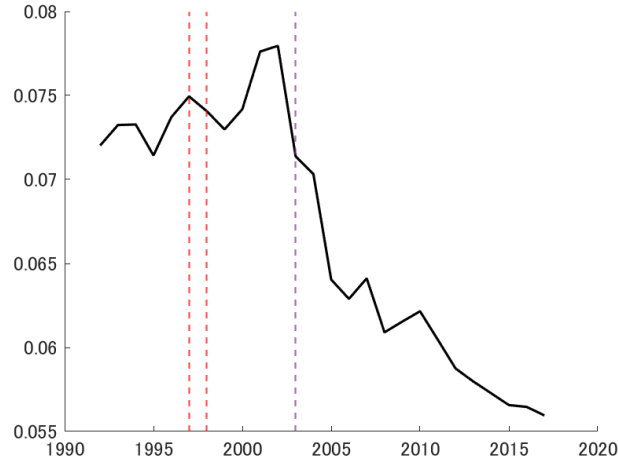
Figure 7 shows the share of outstanding loans from banks that constitute the Resona holdings (Yamato bank, Asahi bank, Kinki-Osaka bank, and Nara bank before the merger) compared

Figure 6: Results for different quartiles



Notes. The figure shows the estimated coefficients on the interaction term between the treatment indicator and the year fixed effect for different quartiles, $\widehat{\beta}_t^q$ for $t = 1990 \dots 2016$ and $q = \{1, 2, 3, 4\}$, using equation 2 when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of the number of employees). The bank bailout occurred in 1997 (*exposure 1*) and in 1998 (*exposure 2*). The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

Figure 7: Share of outstanding loans from Resona Holdings



Notes. The figure shows the share of the outstanding loans from the Resona holdings compared to the overall outstanding loans from all domestic banks in Japan. The dotted vertical lines show the timings of bank bailouts that occurred in 1997, 1998, and 2003.

to the overall outstanding loans from all domestic banks in Japan. Despite the overall declining trend, the share increases steadily leading up to the formation of the Resona holdings in 2003. The data are consistent with the description by [Maeda \(2015\)](#) who documented that the banks that were eventually merged as the Resona holdings tried to expand loans before the merger, especially to small and medium enterprises.

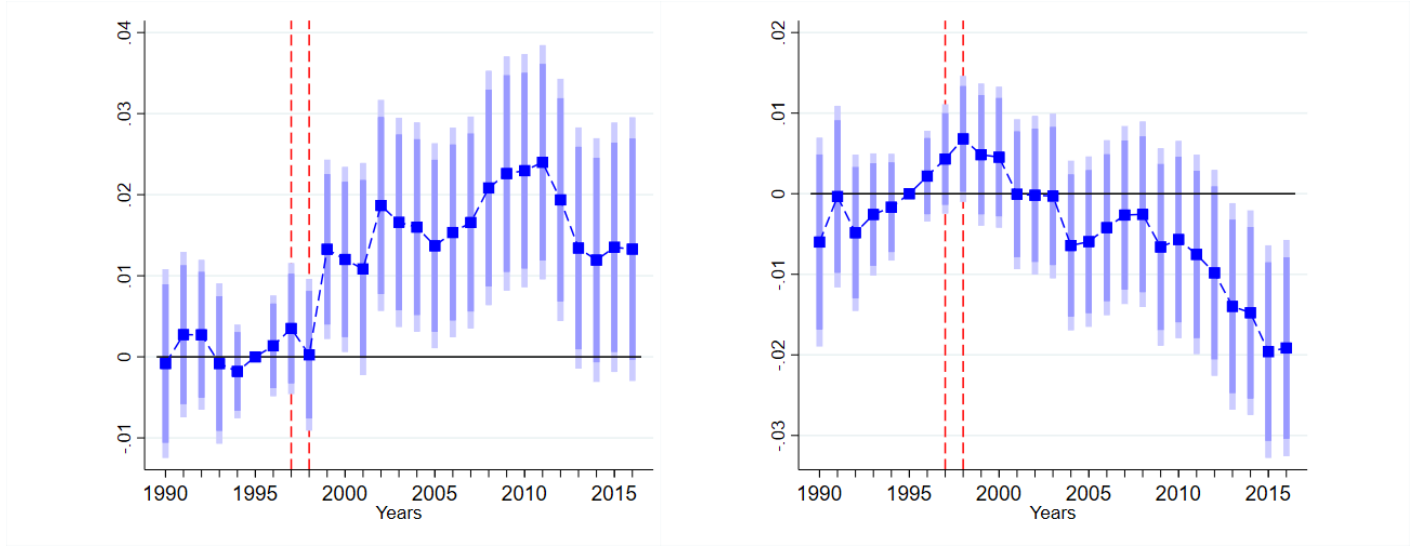
9.3 Additional robustness checks

In this section, I conduct three additional robustness checks, which are the inclusion of market capitalization as an additional control variable, removal of zombie firms from the sample, and use of an alternative definition of the treatment and control groups. The baseline results are robust to all of these variations.

9.3.1 Adding market capitalization as an additional control

First, I include the market capitalization values as a control variable in equation 2 in addition to the other control variables, such as the interest coverage ratio, total liquid liability, firm age and age squared, corporate income tax over total assets, a lagged investment, and a lagged employee growth rate. Figure 8 shows the results. The addition of market capitalization as a control variable increases the standard errors. However, overall, the baseline results are robust to the inclusion of the market capitalization values.

Figure 8: Adding market capitalization as control



(a) Financial debt/total assets

(b) Cash/total assets

Notes. The figure shows $\hat{\beta}_t$ for $t = 1990 \dots 2016$ using equation 2 when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets, log of the number of employees, interest coverage ratio, total liquid liability, firm age and age squared, corporate income tax over total assets, a lagged investment, a lagged employee growth rate, and market capitalization). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

9.3.2 Focusing on non-zombie firms

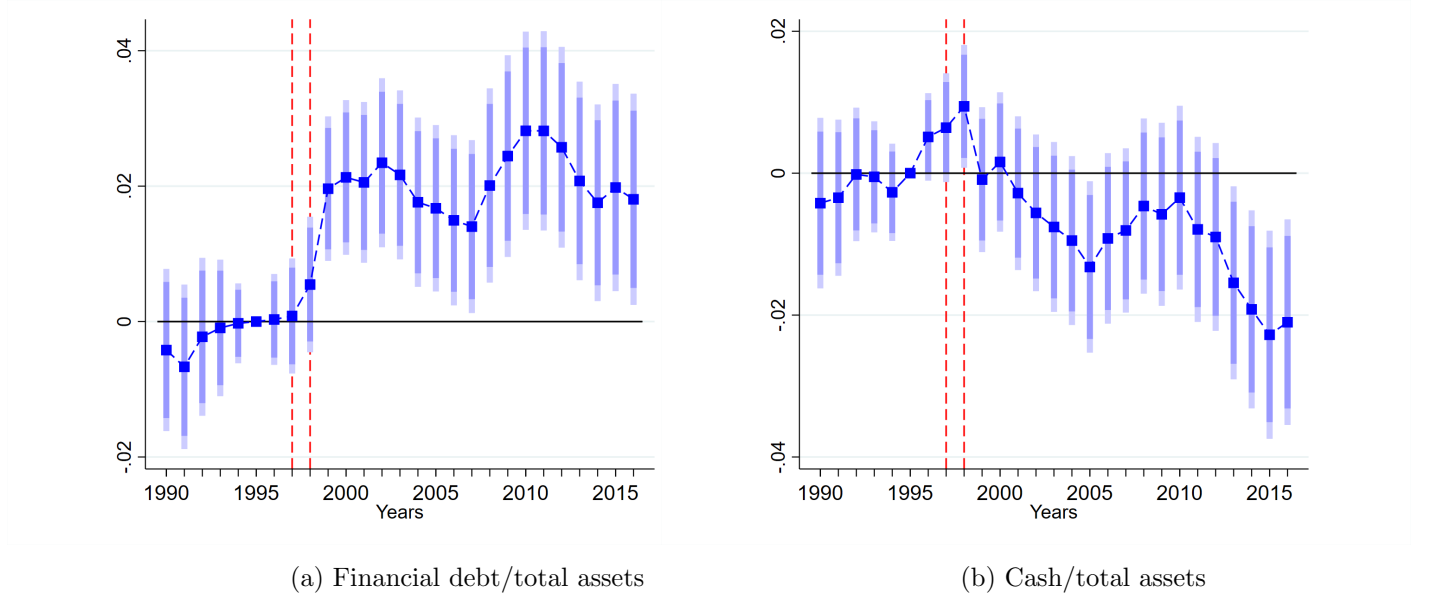
Another concern that could invalidate the baseline result is the existence of zombie firms in the sample. For example, if zombie firms are sampled disproportionately more in the treated group, then the baseline results could be entirely driven by the behavior of zombie firms that could be different from other firms. To check the importance of zombie firms in driving the baseline results, I remove the zombie firms from the sample and re-estimate equation 2. To identify the zombie firms, I follow [Giannetti and Simonov \(2013\)](#) who identify zombie firms in year t as firms whose interest payments in year t were lower than the “required minimum interest rate expenses.” The minimum interest rate expense in year t is calculated by assuming that the borrower pays the average short-term prime rate on short-term bank loans, the average long-term prime rate on long-term bank loans, and zero coupons on total bonds outstanding during year t . [Giannetti and Simonov \(2013\)](#) constructed the zombie indicator for all firms for each year between 1998 and 2004. Using their zombie indicator variable, I remove firms that were identified as zombie firms in any year between 1998 and 2004. The results of estimating equation 2 after removing zombie firms are shown in Figure 9. Again, the baseline results are quantitatively and qualitatively similar to the baseline results.

9.3.3 Using an alternative definition of treatment and control groups

The sample used for the baseline results can be categorized into four groups: 1) firms that were never affected, 2) firms that were affected only by the first round of recapitalization in 1997, 3) firms that were affected only by the second round of recapitalization in 1998, and 4) firms that were affected by both rounds of recapitalization in 1997 and 1998. Our baseline analysis uses the first and second groups as the counterfactual (control) group and the third and fourth groups as the treatment group. In this section, I will examine if changing the definition of the control and treatment groups has any impact on the baseline results.

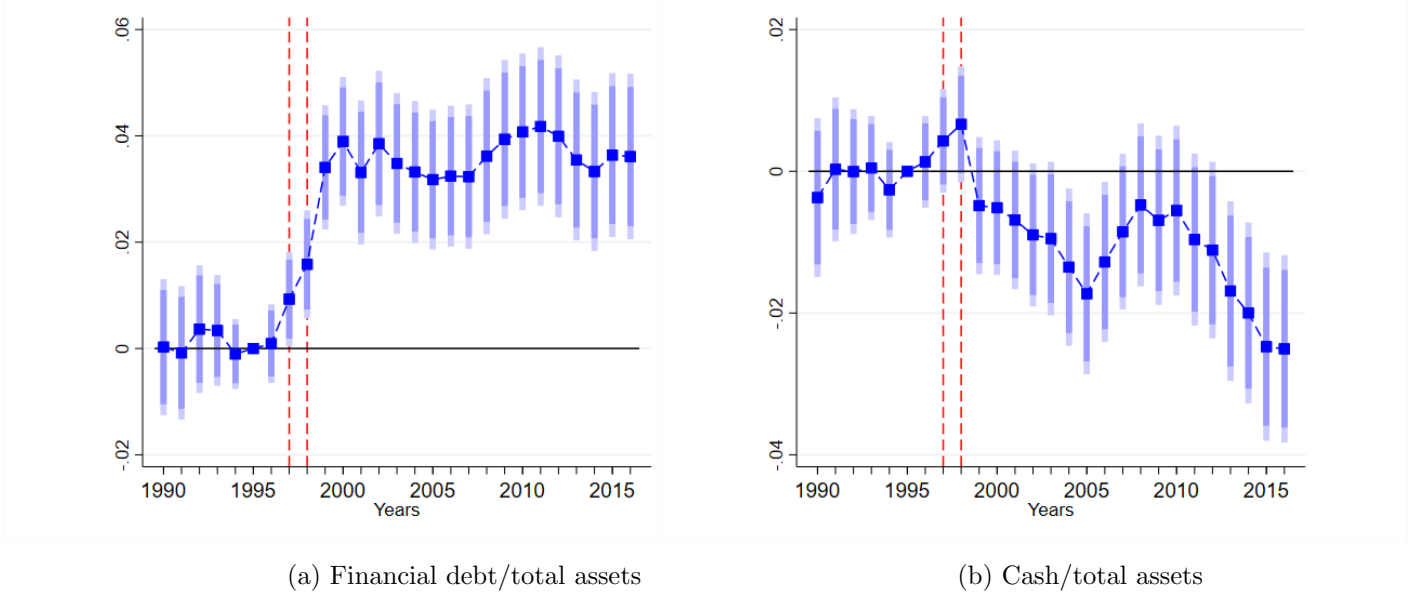
The alternative definition of the control group is the first group, which consists of firms that are never treated. The alternative definition of the treatment group is the fourth group, which are firms that are affected by both the first and second rounds of recapitalization. Thus, I drop the groups of firms that are affected by either the first or second round of recapitalization only. Using this subsample of firms, I re-estimate equation 2. Figure 10 shows the results. The effect of the bailout on the debt over asset ratio slightly increases

Figure 9: Without Zombie firms



Notes. The figure shows the estimated coefficients on the interaction term between the treatment indicator and the year fixed effect for different quartiles, $\hat{\beta}_t$ for $t = 1990 \dots 2016$, using equation 2 when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. The data are restricted to firms that are not identified as zombie firms in any of the periods between 1996 and 2004. Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of the number of employees). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

Figure 10: Alternative definitions of treated and control group:
treated twice v.s. never treated



Notes. The figure shows the estimated coefficients on the interaction term between the treatment indicator and the year fixed effect for different quartiles, $\hat{\beta}_t$ for $t = 1990 \dots 2016$, using equation 2 when the dependent variable is long-term financial debt over total assets in Panel (a) and cash over total assets in Panel (b). The base year is 1995. The data are restricted to firms that are affected by both first and second rounds of recapitalization in 1997 and 19998 (treatment group) and firms that were never affected (control group). Specifications include firm fixed effects, industry-by-year fixed effects, and firm-level covariates (log of total assets and log of the number of employees). The bank bailout occurred in 1997 and in 1998. The dark shaded area represents 90%, and the lightly shaded area 95% firm-cluster-robust confidence intervals, respectively.

compared to the baseline results. Yet, the results are essentially the same as the baseline results. Thus, using the alternative definition of the treatment and control groups does not influence our main findings.

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